

Assignment 3 - Solutions

00103335: Deep Learning and Reinforcement Learning

Fall 2026

Problem 1. Scalar stochastic recursion

Write $h_t = \eta_t$, $A_t = 1 - \lambda h_t \in [0, 1]$, and $v_t = \mathbb{E}|\delta_t|^4$. The initial assumption and $|u + v|^4 \leq 8(|u|^4 + |v|^4)$ show inductively that $v_t < \infty$ for every finite t . Conditional Lyapunov inequalities give

$$\mathbb{E}[\xi_t^2 \mid \mathcal{F}_t] \leq \sigma^2, \quad |\mathbb{E}[\xi_t^3 \mid \mathcal{F}_t]| \leq \sigma^3.$$

- (a) Fix any $c \in (0, 4\lambda)$ and put $\varepsilon = (4\lambda - c)/3 > 0$. Expanding the fourth power conditionally, with $x = \delta_t$ and $h = h_t$, yields

$$\begin{aligned} \mathbb{E}[\delta_{t+1}^4 \mid \mathcal{F}_t] &= A_t^4 x^4 + 6A_t^2 h^2 x^2 \mathbb{E}[\xi_t^2 \mid \mathcal{F}_t] \\ &\quad + 4A_t h^3 x \mathbb{E}[\xi_t^3 \mid \mathcal{F}_t] + h^4 \mathbb{E}[\xi_t^4 \mid \mathcal{F}_t]. \end{aligned}$$

The term involving $x^3 \xi_t$ vanishes because $\mathbb{E}[\xi_t \mid \mathcal{F}_t] = 0$. The cubic-noise term above need not vanish. For $h \leq 1$, Young's inequality and $|x| \leq 1 + x^4$ give

$$\begin{aligned} 6A_t^2 h^2 x^2 \mathbb{E}[\xi_t^2 \mid \mathcal{F}_t] &\leq \varepsilon h x^4 + \frac{9\sigma^4}{\varepsilon} h^3, \\ |4A_t h^3 x \mathbb{E}[\xi_t^3 \mid \mathcal{F}_t]| &\leq 4\sigma^3 h^3 x^4 + 4\sigma^3 h^3, \\ h^4 \mathbb{E}[\xi_t^4 \mid \mathcal{F}_t] &\leq \sigma^4 h^3. \end{aligned}$$

Also, since $0 \leq \lambda h \leq 1$,

$$A_t^4 = (1 - \lambda h)^4 \leq 1 - 4\lambda h + 6\lambda^2 h^2.$$

As $h_t \rightarrow 0$, choose t_1 so that, for $t \geq t_1$, $h_t \leq 1$, $6\lambda^2 h_t^2 \leq \varepsilon$, and $4\sigma^3 h_t^2 \leq \varepsilon$. Combining the bounds gives

$$\mathbb{E}[\delta_{t+1}^4 \mid \mathcal{F}_t] \leq (1 - ch_t)\delta_t^4 + C_0 h_t^3, \quad C_0 := \frac{9\sigma^4}{\varepsilon} + 4\sigma^3 + \sigma^4.$$

Taking expectations proves the claim.

- (b) If $\alpha = 1$, choose c additionally so that $2/a < c < 4\lambda$; this is possible because $2\lambda a > 1$. If $\alpha < 1$, take any $c \in (0, 4\lambda)$. Use the constants t_1, C_0 from part (a) for this c . Set $n = t + t_0$ and

$$D_t := \frac{1 - (h_{t+1}/h_t)^2}{h_t} = \frac{n^\alpha}{a} \left[1 - \left(1 + \frac{1}{n} \right)^{-2\alpha} \right] = \frac{2\alpha}{a} n^{\alpha-1} + O(n^{\alpha-2}).$$

Thus $D_t \rightarrow 0$ when $\alpha < 1$, while $D_t \rightarrow 2/a$ when $\alpha = 1$. In either case choose q with $\lim_{t \rightarrow \infty} D_t < q < c$. Increase $t_2 \geq t_1$ so that $D_t \leq q$ and $ch_t \leq 1$ for all $t \geq t_2$. Choose a finite K satisfying

$$K \geq \frac{v_{t_2}}{h_{t_2}^2}, \quad K(c - q) \geq C_0.$$

If $v_t \leq K h_t^2$ for some $t \geq t_2$, part (a) implies

$$\begin{aligned} v_{t+1} &\leq (1 - ch_t)K h_t^2 + C_0 h_t^3 \\ &\leq K h_t^2 (1 - qh_t) \leq K h_t^2 (1 - D_t h_t) = K h_{t+1}^2. \end{aligned}$$

Induction proves this for all $t \geq t_2$. Enlarging K to cover the finitely many $1 \leq t < t_2$ proves $\mathbb{E}|\delta_t|^4 \leq C \eta_t^2$ for every t .

(c) By part (b), for every $M > 0$ and every t ,

$$\mathbb{P}(|\delta_t| > M\sqrt{\eta_t}) \leq \frac{\mathbb{E}|\delta_t|^4}{M^4\eta_t^2} \leq \frac{C}{M^4}.$$

The right side can be made arbitrarily small uniformly in t ; hence $\delta_t = O_{\mathbb{P}}(\sqrt{\eta_t})$.

(d) Since $2\alpha > 1$,

$$\sum_{t=1}^{\infty} \mathbb{E}|\delta_t|^4 \leq Ca^2 \sum_{t=1}^{\infty} (t+t_0)^{-2\alpha} < \infty.$$

For each $\epsilon > 0$, Markov's inequality therefore gives $\sum_t \mathbb{P}(|\delta_t| > \epsilon) < \infty$. The first Borel–Cantelli lemma, applied to $\epsilon = 1, 1/2, 1/3, \dots$, implies $\delta_t \rightarrow 0$ almost surely.

Problem 2. Convex SGD

Write $e_t = \theta_t - \theta^*$ and $\Delta_t = \Phi(\theta_t) - \Phi(\theta^*) \geq 0$.

(a) Expand the squared distance and condition on $\mathcal{F}_t$:

$$\begin{aligned} \mathbb{E}[\|e_{t+1}\|^2 \mid \mathcal{F}_t] &= \|e_t\|^2 - 2\eta_t \langle e_t, \nabla \Phi(\theta_t) \rangle + \eta_t^2 \mathbb{E}[\|g_t\|^2 \mid \mathcal{F}_t] \\ &\leq \|e_t\|^2 - 2\eta_t \Delta_t + \eta_t^2 G^2. \end{aligned}$$

The last step uses the convex first-order inequality $\langle \nabla \Phi(\theta_t), \theta_t - \theta^* \rangle \geq \Phi(\theta_t) - \Phi(\theta^*)$.

(b) Take expectations in part (a), sum from $t = 1$ to T , and discard the nonnegative terminal squared distance:

$$2 \sum_{t=1}^T \eta_t \mathbb{E} \Delta_t \leq \mathbb{E} \|e_1\|^2 - \mathbb{E} \|e_{T+1}\|^2 + G^2 Q_T \leq R^2 + G^2 Q_T.$$

Because the weights η_t/S_T are nonnegative and sum to one, convexity gives, pathwise,

$$\Phi(\bar{\theta}_T^{(\eta)}) - \Phi(\theta^*) \leq \frac{1}{S_T} \sum_{t=1}^T \eta_t \Delta_t.$$

Taking expectations and using the preceding sum proves

$$\boxed{\mathbb{E}[\Phi(\bar{\theta}_T^{(\eta)})] - \Phi(\theta^*) \leq \frac{R^2 + G^2 Q_T}{2S_T}.$$

(c) With $\eta_t \equiv c/\sqrt{T}$, the weighted average is $\bar{\theta}_T$, while $S_T = c\sqrt{T}$ and $Q_T = c^2$. The bound in part (b) becomes

$$\mathbb{E}[\Phi(\bar{\theta}_T)] - \Phi(\theta^*) \leq \frac{R^2/c + G^2 c}{2\sqrt{T}}.$$

For $R, G > 0$, the right side is minimized at $\boxed{c = R/G}$, yielding the requested $RG/\sqrt{T}$ bound.